perfect **99.13%** model matching accuracy and **89.02%** preference prediction accuracy. Although we compared the captions without the image in this experiment due to cost, the VibeCheck framework can be easily adapted to the multimodal setting.

Prompt: Describe the following image.

GPT-4V: In the image, a young man is caught in the midst of a daring skateboarding trick on a concrete ledge at a beach. His black t-shirt, adorned with a white logo, contrasts with his black shorts and vibrant red shoes. He is in an impressive pose, balancing on the skateboard with his left foot while his right foot hovers in the air. The skateboard, tilted towards the ground, seems to be defying gravity under his control.

The beach forms a lively backdrop to this scene. People can be seen enjoying their time, adding to the dynamic atmosphere. A red and white striped awning stands out, adding a splash of color to the scene. The image captures a moment of thrill and skill, set against the relaxed ambiance of a beach day.

Gemini-1.5-Flash: The image shows a young man skateboarding in a skatepark. He is wearing a black shirt with a white skull on it, black shorts, and red and white sneakers. He is in mid-air, with his skateboard underneath him. The background is a beach, and there are people in the distance. The man is focused on his skateboarding, and he appears to be having fun.

Creative Writing: **GPT-4V:** Contains narrative embellishments and subjective interpretations. **Gemini-1.5-Flash:** Straightforward reporting without creative embellishments
Color and Atmosphere Description: **GPT-4V:** Emphasizes mood and overall ambiance, including colors and lighting. **Gemini-1.5-Flash:** Focuses on identifying and detailing objects without describing the ambiance.
Descriptive Richness: **GPT-4V:** Rich, detailed scene description with story element. **Gemini-1.5-Flash:** Simple, factual description without additional details.

7 Limitations

Although VibeCheck quantifies the impact of each vibe on model identity and user preference, it is challenging to disentangle whether a specific vibe directly influences human preference or if other confounding factors are at play. For example, a model might exhibit a vibe of being more engaging, but its preference by users could stem from its factual accuracy, where accurate outputs incidentally appear more engaging due to their clarity or relevance. Furthermore, the LLM-based vibe discovery process may not capture all relevant differences between models. This is particularly problematic when there's a significant discrepancy in model accuracy, as the discovered vibes may focus primarily on accuracy-related aspects. VibeCheck is also costly to validate, as each judge will have to evaluate each sample in $D_{validation}$ on each vibe. In order for this to be feasible, our method uses relatively inexpensive models such as GPT-4o-mini, but these judge models are often incorrect in their predictions, as shown in Figure 5. LLM judges also have biases (Zheng et al., 2023), like favoring their own outputs, which may affect the scoring. Lastly, running VibeCheck multiple times can lead to different vibes and different results, making it harder to reproduce findings exactly.

8 Conclusion

It may seem unconventional to focus on vibes instead of concrete metrics of correctness, but these qualitative properties have a measurable impact on how people judge models. VibeCheck provides a valuable addition to existing metrics for correctness by capturing these qualitative aspects that influence human preference. As LLM usage expands, we anticipate an increased focus on evaluating vibes to better align with user preferences. Moreover, this approach can be extended to other modalities, such as audio or visual content, and can be applied to compare any pairwise set of texts, making it a versatile tool for model evaluation. In future work, we hope to explore extending this framework to compare a larger number of models along with developing interventions which can use these vibes to improve human preference for given models.

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